

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278539

Kind Code

A1

Publication Date

September 04, 2025

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MODEL CONSTRUCTION FOR SELECTING AQUIFERS FOR CARBON STORAGE

Abstract

An aquifer management system is provided. The aquifer management system includes a computer system and a model manager. The model manager generates simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models. The model manager generates auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters. The model manager creates a training dataset using the simulation data and the auxiliary quantities. The model manager trains a machine learning model using the training dataset to select a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide.

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Family ID: 96881464

Appl. No.: 18/592615

Filed: March 01, 2024

Publication Classification

Int. Cl.: G06F30/27 (20200101); G06F111/10 (20200101)

U.S. Cl.:

CPC G06F30/27 (20200101); G06F2111/10 (20200101)

Background/Summary

BACKGROUND INFORMATION

1. Field

[0001] The present disclosure relates generally to a method and system for selecting aquifers and more specifically to select aquifers with desired capacity and connectivity for storing carbon dioxide.

2. Background

[0002] Aquifers are underground layers for storing and transmitting groundwater. Aquifers usually include porous material such as sand, gravel, or permeable rock that form interconnected spaces to hold water. The geological formation of aquifers can store huge amounts of water and act as natural reservoirs to provide a source of water for wells and springs.

[0003] Further, aquifers can be used to store carbon dioxide to mitigate greenhouse gas emissions. In this case, compressed carbon dioxide can be injected into suitable aquifers, where various trapping techniques can be used to ensure secure storage of carbon dioxide within aquifers.

SUMMARY

[0004] According to one illustrative embodiment, an aquifer management system is provided. The aquifer management system includes a computer system and a model manager. The model manager generates simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models. The model manager generates auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters. The model manager creates a training dataset using the simulation data and the auxiliary quantities. The model manager trains a machine learning model using the training dataset to select a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide.

[0005] According to another illustrative embodiment, a computer implemented method for training a machine learning model to select aquifers for carbon storage is provided. A number of processor units generate simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models. The number of processor units generate auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters. The number of processor units create a training dataset using the simulation data and the auxiliary quantities. The number of processor units. The number of processor units train a machine learning model using the training dataset for selecting a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide.

[0006] According to yet another illustrative embodiment, a computer implemented method for storing carbon dioxide is provided. A number of processor units receive input data for aquifer parameters for a number of candidate aquifers. The number of processor units send the input data to a machine learning model trained using a training dataset comprising combinations of data for the aquifer parameters, simulation parameters, and auxiliary parameters. The number of processor units receive results for candidate aquifers from the machine learning model in response to sending the input data to the machine learning model. The number of processor units identify a set of aquifers that meet a set of carbon storage requirements from the number of candidate aquifers based on the results received from the machine learning model.

[0007] According to yet another illustrative embodiment, a computer program product for training a machine learning model to select aquifers for carbon storage is provided. The computer program product includes a set of one or more computer-readable storage media and program instructions

collectively stored in the set of one or more storage media; cause a number of processor units to generate simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models; generate auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters create a training dataset using the simulation data and the auxiliary quantities; and train a machine learning model using the training dataset to select a number of the aquifers that meet a set of carbon storage requirements to store carbon dioxide.

[0008] The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives, and features thereof will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

[0010] FIG. 1 is a pictorial representation of a network of data processing systems in which illustrative embodiments may be implemented;

[0011] FIG. 2 is a block diagram of an aquifer management environment depicted in accordance with an illustrative embodiment;

[0012] FIG. 3 is a block diagram of an aquifer selection environment depicted in accordance with an illustrative embodiment;

[0013] FIG. 4 depicts a diagram illustrating a user interface in accordance with an illustrative embodiment;

[0014] FIG. 5 depicts a diagram illustrating a user interface in accordance with an illustrative embodiment;

[0015] FIG. 6 depicts a flowchart illustrating a process for training a machine learning model to select aquifers for carbon storage in accordance with an illustrative embodiment;

[0016] FIG. 7 depicts a flowchart illustrating a process for generating a training dataset in accordance with an illustrative embodiment;

[0017] FIG. 8 depicts a flowchart illustrating a process for identifying aquifers to store carbon dioxide in accordance with an illustrative embodiment;

[0018] FIG. 9 depicts a flowchart illustrating a process for identifying aquifers using the machine learning model in accordance with an illustrative embodiment;

[0019] FIG. 10 depicts a flowchart illustrating a process for generating output for selecting aquifers for carbon storage in accordance with an illustrative embodiment;

[0020] FIG. 11 depicts a flowchart illustrating a process for storing carbon dioxide in accordance with an illustrative embodiment;

[0021] FIG. 12 is a block diagram of a data processing system in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

[0022] The illustrative embodiments recognize and take into account a number of considerations. For example, the illustrative embodiments recognize and take into account that geological storage of carbon dioxide includes capture of carbon dioxide, transportation of carbon dioxide, and storage of carbon dioxide.

[0023] The illustrative embodiments also recognize and take into account that standards developed for geological storage of carbon dioxide suggests that storage sites are determined through workflow that start with screening of aquifer sites that begins with an assessment of high level information available for each aquifer site and followed by more detailed characterizations and analysis of the selected sites identified during the screening process.

[0024] The illustrative embodiments also recognize and take into account that current approaches to select aquifers for carbon storages have drawbacks. For example, current approaches include physics-based models that do not properly consider the complex physics of two-phase flow and solubility of the carbon dioxide in the formation brine and cannot accurately account for the interference among wells within an aquifer.

[0025] In another example, current approaches also include physics-based model that use numerical solution to address complex physics issues. However, those models require a large set of input data and therefore are time consuming and require huge amount of computing resources.

[0026] With reference to FIG. 1, a pictorial representation of a network of data processing systems is depicted in which illustrative embodiments may be implemented. Network data processing system **100** is a network of computers in which the illustrative embodiments may be implemented. Network data processing system **100** contains network **102**, which is the medium used to provide communications links between various devices and computers connected together within network data processing system **100**. Network **102** might include connections such as wire, wireless communication links, or fiber optic cables.

[0027] In the depicted example, server computer **104** and server computer **106** connect to network **102** along with storage unit **108**. In addition, client devices **110** connect to network **102**. In the depicted example, server computer **104** provides information such as boot files, operating system images, and applications to client devices **110**. Client devices **110** can be, for example, computers, workstations, or network computers. As depicted, client devices **110** includes client computers **112**, **114**, and **116**. Client devices **110** can also include other types of client devices such as mobile phone **118**, tablet computer **120**, and smart glasses **122**.

[0028] In this illustrative example, server computer **104**, server computer **106**, storage unit **108**, and client devices **110** are network devices that connect to network **102** in which network **102** is the communications media for these network devices. Some or all of client devices **110** may form an Internet of things (IoT) in which these physical devices can connect to network **102** and exchange information with each other over network **102**.

[0029] Client devices **110** are clients to server computer **104** in this example. Network data processing system **100** may include additional server computers, client computers, and other devices not shown. Client devices **110** connect to network **102** utilizing at least one of wired, optical fiber, or wireless connections.

[0030] Program code located in network data processing system **100** can be stored on a computer-recordable storage medium and downloaded to a data processing system or other device for use. For example, the program code can be stored on a computer-recordable storage medium on server computer **104** and downloaded to client devices **110** over network **102** for use on client devices **110**.

[0031] In the depicted example, network data processing system **100** is the Internet with network **102** representing a worldwide collection of networks and gateways that use the Transmission Control Protocol/Internet Protocol (TCP/IP) suite of protocols to communicate with one another. At the heart of the Internet is a backbone of high-speed data communication lines between major nodes or host computers consisting of thousands of commercial, governmental, educational, and other computer systems that route data and messages. Of course, network data processing system **100** also may be implemented using a number of different types of networks. For example, network **102** can be comprised of at least one of the Internet, an intranet, a local area network (LAN), a metropolitan area network (MAN), or a wide area network (WAN). FIG. 1 is intended as an

example and not as an architectural limitation for the different illustrative embodiments.

[0032] In this illustrative example, one or more of client devices **110** can be configured for users and aquifer sites. For example, client computer **114** can include hardware or can be connected to at least one hardware sensors that detect various aquifer parameters for aquifer site **128**. In a similar fashion, client computer **116** can detect various aquifer parameters for aquifer site **130**. In this illustrative example, aquifer site **128** and aquifer site **130** are areas where aquifers can be located.

[0033] For example, client computer **114** and client computer **116** can detect aquifer parameters such as aquifer size, aquifer thickness, aquifer depth, a petrophysical property, a pressure condition, a temperature condition, a fluid property, a solubility condition, a two-phase flow property, a porosity, a fracturing condition, a permeability, or other parameters.

[0034] In this example, user **126** can interact with other devices in network data processing system **100** using one of devices in client devices **110**, such as client computer **112**.

[0035] In this illustrative example, values for aquifer parameters collected by client computer **114** and client computer **116** can be used by model manager **124** on server computer **104** for training a machine learning model to select aquifers for carbon storage.

[0036] In this illustrative example, aquifer selector **132** on server computer **106** can use the machine learning model trained by model manager **124** to select an aquifer that meets a set of carbon storage requirements for carbon storage. In this illustrative example, the set of carbon storage requirements can include capacity, injection rate, a rate of carbon storage, or any parameters that can be used for determining whether an aquifer is suitable for carbon storage.

[0037] For example, user **126** at client computer **112** can provide input data to aquifer selector **132** for selecting aquifer for carbon storage. Input data can be information related to aquifer parameters for a number of candidate aquifers. As a result, aquifer selector **132** uses the input data and the machine learning model trained by model manager **124** to select aquifers from the number of candidate aquifers for carbon storage.

[0038] With reference now to FIG. 2, an illustration of a block diagram of an aquifer management environment is depicted in accordance with an illustrative embodiment. In this illustrative example, aquifer management environment **200** includes components that can be implemented in hardware such as the hardware shown in network data processing system **100** in FIG. 1.

[0039] In this illustrative example, aquifer management system **202** in aquifer management environment **200** uses information received from aquifers **232** to train machine learning models for selecting aquifers that meet carbon storage requirements for carbon storage.

[0040] In this illustrative example, aquifer management system **202** includes computer system **204** and model manager **218**. Model manager **218** is located in computer system **204**. Model manager **218** can be an example of model manager **124** in FIG. 1.

[0041] Model manager **218** can be implemented in software, hardware, firmware, or a combination thereof. When software is used, the operations performed by model manager **218** can be implemented in program instructions configured to run on hardware such as a processor unit. When firmware is used, the operations performed by model manager **218** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit. When hardware is employed, the hardware can include circuits that operate to perform the operations in model manager **218**.

[0042] In the illustrative examples, the hardware can take a form selected from at least one of a circuit system, an integrated circuit, an application specific integrated circuit (ASIC), a programmable logic device, or some other suitable type of hardware configured to perform a number of operations. With a programmable logic device, the device can be configured to perform the number of operations. The device can be reconfigured at a later time or can be permanently configured to perform the number of operations. Programmable logic devices include, for example, a programmable logic array, a programmable array logic, a field programmable logic array, a field programmable gate array, and other suitable hardware devices. Additionally, the processes can be

implemented in organic components integrated with inorganic components and can be comprised entirely of organic components excluding a human being. For example, the processes can be implemented as circuits in organic semiconductors.

[0043] As used herein, “a number of” when used with reference to items means one or more items. For example, “a number of operations” is one or more operations.

[0044] Further, the phrase “at least one of,” when used with a list of items, means different combinations of one or more of the listed items can be used, and only one of each item in the list may be needed. In other words, “at least one of” means any combination of items and number of items may be used from the list, but not all of the items in the list are required. The item can be a particular object, a thing, or a category.

[0045] For example, without limitation, “at least one of item A, item B, or item C,” may include item A, item A and item B, or item B. This example also may include item A, item B, and item C, or item B and item C. Of course, any combination of these items can be present. In some illustrative examples, “at least one of” can be, for example, without limitation, two of item A; one of item B; and ten of item C; four of item B and seven of item C; or other suitable combinations.

[0046] As used herein, when used with reference to items, “a set of” means one or more of the items. For example, “a set of clouds” is one or more different types of cloud environments.

[0047] Computer system **204** is a physical hardware system and includes one or more data processing systems. When more than one data processing system is present in computer system **204**, those data processing systems are in communication with each other using a communications medium. The communications medium can be a network. The data processing systems can be selected from at least one of a computer, a server computer, a tablet computer, or some other suitable data processing system.

[0048] As depicted, computer system **204** includes processor units **216** that are capable of executing program instructions **214** and implementing processes in the illustrative examples. In other words, program instructions **214** are computer-readable program instructions.

[0049] As used herein, a processor unit in processor units **216** is a hardware device and is comprised of hardware circuits such as those on an integrated circuit that respond to and process instructions and program code that operate a computer. When processor units **216** executes program instructions **214** for a process, processor units **216** can be one or more processor units that are in the same computer or in different computers. In other words, the process can be distributed between processor units **216** on the same or different computers in computer system **204**.

[0050] Further, processor units **216** can be of the same type or different types of processor units. For example, processor units **216** can be selected from at least one of a single core processor, a dual-core processor, a multi-processor core, a general-purpose central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or some other type of processor unit.

[0051] In this illustrative example, computer system **204** includes historical data **228** for aquifer parameters **226** collected from aquifers **232**. Historical data **228** includes values for parameters in aquifer parameters **226** collected from aquifers **232** over a period of time. Aquifer parameters **226** are characteristic for aquifers **232**. As depicted, aquifer parameters **226** can include aquifer size, aquifer thickness, aquifer depth, a petrophysical property, a pressure condition, a temperature condition, a fluid property, a solubility condition, a two-phase flow property, a porosity, a fracturing condition, a permeability, or other suitable parameters for aquifers **232**.

[0052] Model manager **218** can use historical data **228** as input to numerical simulation models **224** to generate simulation data **234**. In this illustrative example, historical data **228** can be used as input in a number of ways. For example, model manager **218** can generate a number of input files using different combinations of aquifer parameters in aquifer parameters **226**. Each input file contains values for a combination of aquifer parameters based on historical data **228**.

[0053] In this illustrative example, historical data **228** usually does not have sufficient data for

training machine learning models for selecting aquifers. Therefore, generating simulation data **234** and including simulation data **234** in the training data makes the resulting machine learning model more accurate and efficient.

[0054] As depicted, input files generated using historical data **228** can be used as input for numerical simulation models **224**. Numerical simulation models **224** are representations of real world systems that are used are created using computational tools and techniques. Numerical simulation models **224** are designed to simulate, analyze and predict the behavior of complex systems through mathematical algorithms and numerical simulations.

[0055] For example, numerical simulation models **224** can be numerical simulators that simulate behavior of aquifers **232** based on the input files. In this illustrative example, a simulation is performed for each input file to generate simulation data **234**.

[0056] In this illustrative example, numerical simulation models **224** uses simulation parameters **222** for running simulations. Simulation parameters **222** are characteristic for aquifers **232** that provide information for training machine learning models to select aquifers for carbon storage. For example, simulation parameters **222** can be different combinations of parameters in aquifer parameters **226**.

[0057] Simulation parameters **222** includes selected simulation parameters **252**. Selected simulation parameters **252** can be a subset of simulation parameters **222** and are parameters in simulation parameters **222** that are selected for generating simulation data **234**. In other words, simulation data **234** are values for selected simulation parameters **252**.

[0058] For example, simulation parameters **222** and selected simulation parameters **252** can include at least one of wellbore pressure, reservoir pressure, injection rate, cumulative injection, or combination of parameters. In this example, simulation data **234** can include wellbore pressure and reservoir pressure for a period of time for aquifers **232**, and injection rate and cumulative injection for a period of time for aquifers **232**.

[0059] In another example, simulation data **234** can include wellbore pressure and reservoir pressure in increments of pressure for aquifers **232**, and injection rate and cumulative injection in increments of pressure for aquifers **232**. In this illustrative example, increments of pressure are values of pressure between the initial aquifer pressure and the fracture pressure. In other words, wellbore pressure, reservoir pressure, injection rate, and cumulative injection can be calculated at increments of pressure.

[0060] In this illustrative example, model manager **218** uses simulation data **234** to generate auxiliary quantities **238**. Auxiliary quantities **238** are values for auxiliary parameters **242**. Auxiliary parameters **242** are variables that represent deviation between a simplified model and reality. Auxiliary parameters **242** can also be referred to as “skin term”.

[0061] In this illustrative example, the simplified model is a mathematical relationship that employs simplified assumptions to represent behaviors for aquifers **232**. For example, the simplified model can be mathematical relationships from physics-based equations **240** that do not take auxiliary parameters **242** into account. In other words, the simplified model is a mathematical relationship that does not take auxiliary parameters **242** into account and therefore cannot generate accurate predictions to forecast behavior for aquifers **232**.

[0062] In this illustrative example, auxiliary parameters **242** can be positive or negative, with the absolute values of auxiliary parameters **242** indicate the extent of deviation. In other words, auxiliary parameters **242** can be included as part of the training data to improve accuracy of machine learning model.

[0063] In this illustrative example, auxiliary quantities **238** for auxiliary parameters **242** can be generated using physics-based equations **240**. Physics-based equations **240** are reservoir engineering equations. In this illustrative example, reservoir engineering equations are mathematical relationships used to analyze and model the behavior of fluids within underground reservoirs.

[0064] In this illustrative example, physics-based equations **240** include a material balance equation that defines relation between average reservoir pressure and cumulative carbon dioxide injected into an aquifer. The material balance equation is as follows:

$$p(t) - p_{\text{sub.i}} = B_{\text{sub.g}} \int q(t) dt / A h \emptyset c_{\text{sub.t}} \quad (1)$$

Where $p(t)$ is the average reservoir pressure, $q(t)$ is the injection rate, $\int q(t) dt$ is the cumulative value of the injected carbon dioxide, A is the reservoir area, h is the reservoir thickness, $\emptyset$ is the porosity, $B_{\text{sub.g}}$ is the gas formation volume factor, $p_{\text{sub.i}}$ is initial reservoir pressure, and $c_{\text{sub.t}}$ is the total compressibility of rock and fluid in aquifer.

[0065] In this illustrative example, compression refers to the reduction in volume of a substance when pressure is applied, and compressibility refers to how much the rock and fluid can be squeezed when pressure is applied.

[0066] Material balance equation provides a relationship that is crucial for selecting aquifers for carbon storage. In this illustrative example, the relationship between average reservoir pressure and cumulative volume of carbon dioxide injected is one of the key features for selecting aquifers for carbon storage because the cumulative volume of carbon dioxide injected is the total amount of gas that can be injected into aquifers, and average reservoir pressure is the limit to which capacity is determined.

[0067] In this illustrative example, the material balance equation is designed for water injection into a water reservoir. However, such an assumption can be corrected by inputting simulation data **234** into physics-based equations **240**. In other words, the material balance equation can properly model carbon dioxide injection into an aquifer to provide an accurate relationship between average reservoir pressure and cumulative carbon dioxide injection upon using simulation data **234** as inputs.

[0068] In addition, physics-based equations **240** can also include a deliverability equation that defines the relationship between injection rate and difference between average reservoir pressure and wellbore pressure for the aquifers. The deliverability equation is as follows:

$$[00001] \quad q(t) = \frac{2}{B_g} \frac{k h}{\mu_w} \frac{p_w(t) - \bar{p}(t)}{\ln(r_e/r_w) - 0.75 + S(t)} \quad (2)$$

where $p_{\text{sub.w}}(t)$ is the maximum injection pressure, $q(t)$ is the injection rate, $\mu_{\text{sub.w}}$ is the water viscosity, $r_{\text{sub.e}}$ is the external radius of aquifers, $r_{\text{sub.w}}$ is the wellbore radius for aquifers, k is the permeability of aquifers, h is the thickness for aquifers, $B_{\text{sub.g}}$ is the gas formation volume factor, $p(t)$ is the average reservoir pressure, and $S(t)$ is the value for skin term that can be used as auxiliary quantities **238** for auxiliary parameters **242**.

[0069] In this illustrative example, simulation data **234** and auxiliary quantities **238** are used to form training dataset **236**. In another example, training dataset **236** can also include historical data **228** for aquifer parameters **226** as part of training dataset **236**.

[0070] Computer system **204** further includes machine intelligence **212**. Machine intelligence **212** includes machine learning models **248** and machine learning algorithms **250**. Machine learning is a branch of artificial intelligence (AI) that enables computers to detect patterns and improve performance without direct programming commands. Rather than relying on direct input commands to complete a task, machine learning relies on input data. The data is fed into the machine, one of machine learning algorithms **250** is selected, parameters for the data are configured, and the machine is instructed to find patterns in the input data through optimization algorithms. The data model formed from analyzing the data is then used to predict future values. In this illustrative example, the learning of the machine learning models **248** can be achieved through a database input that is continuously refined over time through trial and error. Equivalence of assets or products can be effectively performed by supervised machine learning so that products or assets that do not match descriptively can nevertheless be matched. Over time, the data model from machine learning can provide a greater degree of flexibility in matching for machine learning

models **248**.

[0071] In addition, machine intelligence **212** can also include deep learning and deep learning algorithms. Deep learning is a method of artificial intelligence that mimics the human brain's capacity to learn and adapt. Deep learning utilizes neural networks that have multiple layers for identifying and learning features from data. In this illustrative example, deep learning can use an iterative process such as backpropagation and gradient descent to refine its parameters to make accurate predictions by minimizing the difference between outputs and actual results.

[0072] Machine intelligence **212** can be implemented using one or more systems such as an artificial intelligence system, a neural network, a generative adversarial network, a generative neural network, a Bayesian network, an expert system, a fuzzy logic system, a genetic algorithm, or other suitable types of systems. In this illustrative example, machine learning models **248** and machine learning algorithms **250** may make computer system **204** a special purpose computer for generating machine learning models for selecting aquifers for carbon storage.

[0073] Machine learning models **248** involves using machine learning algorithms **250** to build machine intelligence **212** based on samples of data such as training dataset **236**. In other words, machine learning models **248** are trained using training dataset **236**.

[0074] As depicted, machine intelligence **212** includes machine learning models **248** that can be trained using training dataset **236**. Machine intelligence **212** can be used to make predictions without being explicitly programmed to make these predictions and can be trained and retrained for a number of different types of applications. These applications include, for example, medicine, financial services, healthcare, speech recognition, computer vision, or other types of applications.

[0075] Machine learning algorithms **250** can include supervised machine learning algorithms, unsupervised machine learning algorithms, and self-learning algorithms. Supervised machine learning can train machine learning models using data containing both the inputs and desired outputs. Examples of machine learning algorithms include XGBoost, K-means clustering, and random forest.

[0076] In addition, SHapley Additive explanation (SHAP) tool can be used to confirm that machine learning models **248** accurately captures effect of various parameters. In other words, SHapley Additive explanation tool can be used to verify the reliability of machine learning model **248**.

[0077] In this illustrative example, machine learning models **248** can be used to select aquifers that meet a set of requirements for storing carbon dioxide. For example, machine learning models **248** can be used to select set of aquifers **230** that meet the requirements from carbon storage requirements **220**. Carbon storage requirements **220** define conditions for storing carbon dioxide in underground geological formations. In this illustrative example, an aquifer can be a type of underground carbon storage.

[0078] In this illustrative example, carbon storage requirements **220** can include capacity, injection rate, a rate of carbon storage, or any parameters and characteristics for aquifers that can be used for determining whether an aquifer is suitable for carbon storage.

[0079] In this illustrative example, the set of aquifers **230** can be aquifers at a single storage site that meets requirements in carbon storage requirements **220**. In another illustrative example, the set of aquifers **230** can be a ranked list of single storage sites that meets requirements in carbon storage requirements **220**. In yet another illustrative example, the set of aquifers **230** can be multiple aquifers that meet the requirements in carbon storage requirements **220**.

[0080] In this illustrative example, user **206** can interact with computer system **204** to coordinate generation of machine learning models **248**. User **206** can be an example of user **126** in FIG. 1. Computer system **204** can receive a user input **208** from user **206**.

[0081] In this example, user input **208** can be generated by user **206** using human machine interface (HMI) **210**. As depicted, human machine interface **210** includes display system **244** and input system **246**. Display system **244** is a physical hardware system and includes one or more display devices on which graphical user interface **256** can be displayed. The display devices can

include at least one of a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a computer monitor, a projector, a flat panel display, a heads-up display (HUD), a head-mounted display (HMD), smart glasses, augmented reality glasses, or some other suitable device that can output information for the visual presentation of information.

[0082] In this illustrative example, graphical user interface **256** can be used to display information to user **206**. For example, graphical user interface **256** can display simulation data **234**, historical data **228**, analysis for aquifers **232** and set of aquifers **230**, or any information associated with aquifers **232** and set of aquifers **230**.

[0083] In this example, user **206** is a person that can interact with graphical user interface **256** through user input **208** generated by input system **246**. Input system **246** is a physical hardware system and can be selected from at least one of a mouse, a keyboard, a touch pad, a trackball, a touchscreen, a stylus, a motion sensing input device, a gesture detection device, a data glove, a cyber glove a haptic feedback device, or some other suitable type of input device.

[0084] In this illustrative example, user **206** can review information associated with set of aquifers **230** to determine accuracy of machine learning models **248** using graphical user interface **256**. For example, user **206** can review information associated with set of aquifers **230** to determine whether set of aquifers **230** meet the requirements from carbon storage requirements **220**. After reviewing, user **206** can provide feedback **254** through user input **208** to model manager **218**.

[0085] Model manager **218** can also use feedback **254** to perform retraining of machine learning models **248** to improve machine intelligence **212**. In this illustrative example, machine learning algorithms **250** in machine intelligence **212** can use feedback from feedback **254** in user input **208** received from user **206** to retrain machine learning models **248** such that accuracy and quality of prediction generated by machine learning models **248** can be improved over time.

[0086] In one illustrative example, one or more solutions are present that overcome a problem with constructing machine learning models for selecting aquifers for carbon storage. As a result, one or more technical solutions may provide an ability to increase efficiency and accuracy in selecting for carbon storage. Thus, the inconveniences and errors from manually updating source code in different versions can be reduced.

[0087] In the illustrative example, computer system **204** can be configured to perform at least one of the steps, operations, or actions described in the different illustrative examples using software, hardware, firmware, or a combination thereof. As a result, computer system **204** operates as a special purpose computer system in which model manager **218** in computer system **204** enables managing the execution of generating training dataset **236**, training of machine learning models **248**, assessing performance for machine learning models **248**, and retraining of machine learning models **248**. In particular, model manager **218** transforms computer system **204** into a special purpose computer system as compared to currently available general computer systems that do not have model manager **218**.

[0088] In the illustrative example, the use of model manager **218** in computer system **204** integrates processes into a practical application for generating training data and constructing computation model using the training data to increases the performance of computer system **204**. In other words, model manager **218** in computer system **204** is directed to a practical application of processes integrated into model manager **218** in computer system **204** that constructs machine learning models for selecting aquifers for carbon storage.

[0089] The illustration of aquifer management environment **200** in FIG. 2 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment can be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment. For example, set of aquifers **230** and aquifers **232** can share at least a portion of aquifers.

[0090] In another example, model manager **218** can also be configured to select multi-well aquifers when single-well aquifers are not sufficient for carbon storage. In this illustrative example, selected simulation parameters **252** can include “number of wells” and “well arrangement” such as spacing between wells such that simulation data **234** is also accounted for multi-well situations.

[0091] With reference now to FIG. **3**, an illustration of a block diagram of an aquifer selection environment is depicted in accordance with an illustrative embodiment. In this illustrative example, aquifer selection environment **300** includes components that can be implemented in hardware such as the hardware shown in network data processing system **100** in FIG. **1**.

[0092] In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures.

[0093] In this illustrative example, aquifer selection system **302** in aquifer selection environment **300** uses information received from candidate aquifers **304** to select set of aquifers **230** that meet the requirements in carbon storage requirements **220** for carbon storage.

[0094] In this illustrative example, aquifer selection system **302** includes computer system **204** and aquifer selector **306**. Aquifer selector **306** is located in computer system **204**. Aquifer selector **306** can be an example of aquifer selector **132** in FIG. **1**.

[0095] Aquifer selector **306** can be implemented in software, hardware, firmware, or a combination thereof. When software is used, the operations performed by aquifer selector **306** can be implemented in program instructions configured to run on hardware, such as a processor unit. When firmware is used, the operations performed by aquifer selector **306** can be implemented in program instructions and data and stored in persistent memory to run on a processor unit. When hardware is employed, the hardware can include circuits that operate to perform the operations in aquifer selector **306**.

[0096] As depicted, machine intelligence **212** includes machine learning models **248** trained using training dataset **236**. In this illustrative example, aquifer selector **306** receives input data **314** for aquifer parameters **226**. Input data **314** are values for aquifer parameters **226** for candidate aquifers **304**.

[0097] In this illustrative example, aquifer selector **306** uses machine learning models **248** to determine auxiliary quantities **312** based on auxiliary parameters **242** from input data **314** for aquifer parameters **226**. In this illustrative example, aquifer selector **306** can use machine learning models **248** with auxiliary quantities **312** to determine values **310** for carbon storage parameters **308**. Carbon storage parameters **308** are parameters that can be used for determining whether an aquifer in candidate aquifers **304** is suitable for carbon storage. For example, carbon storage parameters **308** can include capacity, injection rate, or a rate of carbon storage.

[0098] Aquifer selector **306** compares values **310** to carbon storage requirements **220** to form comparisons **318**. In other words, comparisons **318** provides indications on whether aquifers in candidate aquifers **304** meet requirements in carbon storage requirements **220** for carbon storage.

[0099] In this illustrative example, aquifer selector **306** selects set of aquifers **230** in candidate aquifers **304** based on comparisons **318**. In other words, set of aquifers **230** are aquifers from candidate aquifers **304** that meet the requirements in carbon storage requirements **220** for carbon storage. In this illustrative example, carbon dioxide can be stored in set of aquifers **230** after selection for set of aquifers **230** is made by aquifer selector **306**.

[0100] In this illustrative example, aquifer selector **306** can also generate results **316** for set of aquifers **230** using machine learning models **248** based on input data **314** for aquifer parameters **226**. Results **316** can include analysis and decisions associated with carbon storage for set of aquifers **230**. For example, results **316** can include an order in which carbon dioxide is to be stored in set of aquifers **230**. In another example, results **316** can include a rate at which carbon dioxide is to be stored in set of aquifers **230**.

[0101] In one illustrative example, one or more solutions are present that overcome a problem with

selecting aquifers for carbon storage. As a result, one or more technical solutions may provide an ability to increase efficiency and accuracy in selecting aquifers for carbon storage. Thus, the inconveniences and errors from manually updating source code in different versions can be reduced.

[0102] In a similar fashion, computer system **204** operates as a special purpose computer system in which aquifer selector **306** in computer system **204** enables managing the execution of selecting set of aquifers **230** that meet requirements in carbon storage requirements **220** using machine learning models **248** and generate results **316** for set of aquifers **230** using machine learning models **248**. In particular, aquifer selector **306** transforms computer system **204** into a special purpose computer system as compared to currently available general computer systems that do not have aquifer selector **306**.

[0103] In a similar fashion, the use of aquifer selector **306** in computer system **204** integrates processes into a practical application for selecting aquifers for carbon storage. In other words, aquifer selector **306** in computer system **204** is directed to a practical application of processes integrated into aquifer selector **306** in computer system **204** that selects aquifers for carbon storage.

[0104] The illustration of aquifer selection environment **300** in FIG. 3 is not meant to imply physical or architectural limitations to the manner in which an illustrative embodiment can be implemented. Other components in addition to or in place of the ones illustrated may be used. Some components may be unnecessary. Also, the blocks are presented to illustrate some functional components. One or more of these blocks may be combined, divided, or combined and divided into different blocks when implemented in an illustrative embodiment. For example, comparisons **318** can be generated as part of results **316** using machine learning models **248**.

[0105] In another example, aquifer selector **306** can also take incomplete data or uncertainties from input data **314** into account. In this illustrative example, aquifer selector **306** can use default value for aquifer parameters in incomplete data from input data **314** and Monte-Carlo simulation to estimate range of values to account for uncertainties in input data **314**.

[0106] FIG. 4 depicts a diagram illustrating a user interface in accordance with an illustrative embodiment. In this illustrative example, the user interface illustrated in FIG. 4 can be an example of graphical user interface **256** displayed using display system **244** in

[0107] FIG. 2.

[0108] In FIG. 4, an analysis of a single aquifer is displayed in user interface **400**. User interface **400** is divided into different sections for presenting analysis for an aquifer. In this illustrative example, input area **402** presents a portion of user interface **400** for inputting aquifer parameters for the aquifer.

[0109] In addition, a number of plots can be presented in user interface **400**. For example, Plot **404** presents increment of pressure in relation to capacity of the aquifer for the aquifer parameters input through input area **402**. Plot **404** is colored by density of data points representing number of samples provided in input area **402**.

[0110] In this illustrative example, the color plot can be considered as visual guides displaying how the numerical data, grouped into intervals, spreads across both the horizontal (x) and vertical (y) axes. The color plot can provide a clear picture of the probable distribution of data. For example, regions exhibiting a density higher than the median are represented by various colors, indicating the most probable areas for generated samples. The max increment of pressure line on plot **404** depicts the value input through input area **402**.

[0111] Plot **406** presents capacity distribution of aquifer at maximum allowable reservoir pressure for the aquifer parameters input through input area **402**. In this example, plot **406** presents detailed distribution of capacities on max increment of pressure line on plot **404**.

[0112] Plot **408** presents single well injection rate of the aquifer over time for the aquifer parameters input through input area **402**. Plot **408** is colored based on the density of data points, providing a visual representation of the most probable injection rates for generated samples. The

Contract Injection Rate (Mt/year) line on plot **408** depicts the value input through input area **402**. [0113] In addition, Plot **410** presents a detailed distribution of injection rates at early times for the number of samples provided in input area **402**.

[0114] It should be understood that the presentation of plots in FIG. **4** is only one embodiment of the present disclosure. For example, other types of analysis can also be displayed using user interface **400**.

[0115] FIG. **5** depicts a diagram illustrating a user interface in accordance with an illustrative embodiment. In this illustrative example, the user interface illustrated in FIG. **5** can be an example of graphical user interface **256** displayed using display system **244** in FIG. **2**.

[0116] In the illustrative examples, the same reference numeral may be used in more than one figure. This reuse of a reference numeral in different figures represents the same element in the different figures.

[0117] In FIG. **5**, an analysis for multi-well aquifer is displayed in user interface **500**. In a similar fashion, user interface **500** is divided into different sections for presenting analysis for a multi-well aquifer.

[0118] A number of plots can be presented in user interface **500**. For example, plot **502** presents injection rate compliance matrix for a multi-well aquifer. The injection rate compliance matrix displays potential configuration for the number of wells and well spacing based on input values provided in input area **402** in FIG. **4**. The different colors indicate cells that either meet or do not meet the specified contract injection rate value provided in input area **402**. In this illustrative example, user interface **500** can be interactive such that a click on any cell within the matrix to see the corresponding relevant plots such as plot **504**, **506**, **508**, and **510**.

[0119] Plot **504** presents increments of pressure over time for number of samples provided in input area **402** in FIG. **4**. Plot **504** is colored based on density of data points. The max increment of pressure line on plot **504** depicts the value input through input area **402**.

[0120] Plot **506** presents capacity over time for the number of samples provided in input area **402** in FIG. **4**. In a similar fashion, plot **506** is colored based on density of data points. In plot **506**, the project duration line on plot **506** depicts the value input through input area **402**.

[0121] Plot **508** presents multi-well injection rate over time for the number of samples provided in input area **402**. Plot **508** is colored based on density of data points.

[0122] Plot **510** presents a detailed distribution of multi-well injection rates at early times for number of samples provided in input area **402** in FIG. **4**.

[0123] It should also be understood that the presentation of plots in FIG. **5** is only one embodiment of the present disclosure. For example, other types of analysis can also be displayed using user interface **500**.

[0124] FIG. **6** depicts a flowchart illustrating a process for training a machine learning model to select aquifers for carbon storage in accordance with an illustrative embodiment. The process in FIG. **6** can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that are run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in model manager **218** in computer system **204** in FIG. **2**.

[0125] The process begins by generating simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models (step **600**).

[0126] The process generates auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters (step **602**). The process creates a training dataset using the simulation data and the auxiliary quantities (step **604**).

[0127] The process trains a machine learning model using the training dataset for selecting a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide (step **606**). The

process terminates thereafter.

[0128] Turning next to FIG. 7, a flowchart of a process for generating a training dataset is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for step **604** in FIG. 6.

[0129] The process begins by generating the training dataset using combinations of data for the aquifer parameters, simulation parameters, and the auxiliary parameters (step **700**). The process terminates thereafter.

[0130] Turning next to FIG. 8, a flowchart of a process for identifying aquifers to store carbon dioxide is depicted in accordance with an illustrative embodiment. The process in this figure is an example of an additional step that can be performed with the steps in FIG. 6.

[0131] The process begins by receiving input data for the aquifer parameters for a number of candidate aquifers (step **800**). The process identifies the set of aquifers that meet the set of carbon storage requirements for carbon storage from the number of candidate aquifers using the machine learning model (step **802**). The process terminates thereafter.

[0132] Turning next to FIG. 9, a flowchart of a process for identifying aquifers using the machine learning model is depicted in accordance with an illustrative embodiment. The process in this flowchart is an example of an implementation for step **802** in FIG. 8.

[0133] The process begins by determining the auxiliary quantities for the auxiliary parameters from the input data for the aquifer parameters using the machine learning model (step **900**). The process determines values for carbon storage parameters using the auxiliary quantities for the auxiliary parameters (step **902**). The process compares the values for the carbon storage parameters with the set of carbon storage requirements to form a comparison (step **904**). The process identifies the set of aquifers that meet the set of carbon storage requirements from the number of candidate aquifers based on the comparison (step **906**). The process terminates thereafter.

[0134] Turning next to FIG. 10, a flowchart of a process for generating output for selecting aquifers for carbon storage is depicted in accordance with an illustrative embodiment. The process in this figure is an example of an additional step that can be performed with the steps in FIG. 9.

[0135] The process begins by determining at least one of a range of capacity and injection rate, an injection rate profile, a number of injectors, or a well spacing for the number of the aquifers (step **1000**). The process terminates thereafter.

[0136] Turning next to FIG. 11, a flowchart of a process for storing carbon dioxide is depicted in accordance with an illustrative embodiment. The process in FIG. 11 can be implemented in hardware, software, or both. When implemented in software, the process can take the form of program instructions that are run by one of more processor units located in one or more hardware devices in one or more computer systems. For example, the process can be implemented in aquifer selector **306** in computer system **204** in FIG. 3.

[0137] The process begins by receiving input data for aquifer parameters for a number of candidate aquifers (step **1100**). The process sends the input data to a machine learning model trained using a training dataset comprising combinations of data for the aquifer parameters, simulation parameters, and auxiliary parameters (step **1102**). The process receives results for candidate aquifers from the machine learning model in response to sending the input data to the machine learning model (step **1104**). The process identifies a set of aquifers that meet a set of carbon storage requirements from the number of candidate aquifers based on the results received from the machine learning model (step **1106**). The process terminates thereafter.

[0138] The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams may represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program instructions, hardware, or a combination of the program instructions and hardware. When implemented in

hardware, the hardware may, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program instructions and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams can be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program instructions run by the special purpose hardware.

[0139] In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession can be performed substantially concurrently, or the blocks may sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks can be added in addition to the illustrated blocks in a flowchart or block diagram.

[0140] Turning now to FIG. 12, an illustration of a block diagram of a data processing system is depicted in accordance with an illustrative embodiment. Data processing system **1200** may be used to implement server computer **104** and server computer **106** and client devices **110** in FIG. 1, as well as computer system **204** in FIG. 2 and FIG. 3. In this illustrative example, data processing system **1200** includes communications framework **1202**, which provides communications between processor unit **1204**, memory **1206**, persistent storage **1208**, communications unit **1210**, input/output unit **1212**, and display **1214**. In this example, communications framework **1202** may take the form of a bus system.

[0141] Processor unit **1204** serves to execute instructions for software that may be loaded into memory **1206**. Processor unit **1204** may be a number of processors, a multi-processor core, or some other type of processor, depending on the particular implementation. In an embodiment, processor unit **1204** comprises one or more conventional general-purpose central processing units (CPUs). In an alternate embodiment, processor unit **1204** comprises one or more graphical processing units (GPUS).

[0142] Memory **1206** and persistent storage **1208** are examples of storage devices **1216**. A storage device is any piece of hardware that is capable of storing information, such as, for example, without limitation, at least one of data, program code in functional form, or other suitable information either on a temporary basis, a permanent basis, or both on a temporary basis and a permanent basis. Storage devices **1216** may also be referred to as computer-readable storage devices in these illustrative examples. Memory **1206**, in these examples, may be, for example, a random access memory or any other suitable volatile or non-volatile storage device. Persistent storage **1208** may take various forms, depending on the particular implementation.

[0143] For example, persistent storage **1208** may contain one or more components or devices. For example, persistent storage **1208** may be a hard drive, a flash memory, a rewritable optical disk, a rewritable magnetic tape, or some combination of the above. The media used by persistent storage **1208** also may be removable. For example, a removable hard drive may be used for persistent storage **1208**. Communications unit **1210**, in these illustrative examples, provides for communications with other data processing systems or devices. In these illustrative examples, communications unit **1210** is a network interface card.

[0144] Input/output unit **1212** allows for input and output of data with other devices that may be connected to data processing system **1200**. For example, input/output unit **1212** may provide a connection for user input through at least one of a keyboard, a mouse, or some other suitable input device. Further, input/output unit **1212** may send output to a printer. Display **1214** provides a mechanism to display information to a user.

[0145] Instructions for at least one of the operating system, applications, or programs may be located in storage devices **1216**, which are in communication with processor unit **1204** through communications framework **1202**. The processes of the different embodiments may be performed by processor unit **1204** using computer-implemented instructions, which may be located in a

memory, such as memory **1206**.

[0146] These instructions are referred to as program code, computer-usable program code, or computer-readable program code that may be read and executed by a processor in processor unit **1204**. The program code in the different embodiments may be embodied on different physical or computer-readable storage media, such as memory **1206** or persistent storage **1208**.

[0147] Program code **1218** is located in a functional form on computer-readable media **1220** that is selectively removable and may be loaded onto or transferred to data processing system **1200** for execution by processor unit **1204**. Program code **1218** and computer-readable media **1220** form computer program product **1222** in these illustrative examples. In one example, computer-readable media **1220** may be computer-readable storage media **1224** or computer-readable signal media **1226**.

[0148] In these illustrative examples, computer-readable storage media **1224** is a physical or tangible storage device used to store program code **1218** rather than a medium that propagates or transmits program code **1218**. Computer-readable storage media **1224**, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0149] Alternatively, program code **1218** may be transferred to data processing system **1200** using computer-readable signal media **1226**. Computer-readable signal media **1226** may be, for example, a propagated data signal containing program code **1218**. For example, computer-readable signal media **1226** may be at least one of an electromagnetic signal, an optical signal, or any other suitable type of signal. These signals may be transmitted over at least one of communications links, such as wireless communications links, optical fiber cable, coaxial cable, a wire, or any other suitable type of communications link.

[0150] The different components illustrated for data processing system **1200** are not meant to provide architectural limitations to the manner in which different embodiments may be implemented. The different illustrative embodiments may be implemented in a data processing system including components in addition to or in place of those illustrated for data processing system **1200**. Other components shown in FIG. **12** can be varied from the illustrative examples shown. The different embodiments may be implemented using any hardware device or system capable of running program code **1218**.

[0151] The flowcharts and block diagrams in the different depicted embodiments illustrate the architecture, functionality, and operation of some possible implementations of apparatuses and methods in an illustrative embodiment. In this regard, each block in the flowcharts or block diagrams can represent at least one of a module, a segment, a function, or a portion of an operation or step. For example, one or more of the blocks can be implemented as program code, hardware, or a combination of the program code and hardware. When implemented in hardware, the hardware may, for example, take the form of integrated circuits that are manufactured or configured to perform one or more operations in the flowcharts or block diagrams. When implemented as a combination of program code and hardware, the implementation may take the form of firmware. Each block in the flowcharts or the block diagrams may be implemented using special purpose hardware systems that perform the different operations or combinations of special purpose hardware and program code run by the special purpose hardware.

[0152] In some alternative implementations of an illustrative embodiment, the function or functions noted in the blocks may occur out of the order noted in the figures. For example, in some cases, two blocks shown in succession may be performed substantially concurrently, or the blocks may

sometimes be performed in the reverse order, depending upon the functionality involved. Also, other blocks may be added in addition to the illustrated blocks in a flowchart or block diagram. [0153] The different illustrative examples describe components that perform actions or operations. In an illustrative embodiment, a component may be configured to perform the action or operation described. For example, the component may have a configuration or design for a structure that provides the component with an ability to perform the action or operation that is described in the illustrative examples as being performed by the component.

[0154] Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other illustrative embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. An aquifer management system comprising: a computer system; a model manager configured to: generate simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models; generate auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters; create a training dataset using the simulation data and the auxiliary quantities; and train a machine learning model using the training dataset to select a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide.
2. The aquifer management system of claim 1, wherein creating the training dataset using the simulation data and the auxiliary quantities, the model manager is configured to: generate the training dataset using combinations of data for the aquifer parameters, simulation parameters, and the auxiliary parameters.
3. The aquifer management system of claim 1, wherein the set of carbon storage requirement are selected from at least one of capacity, injection rate, or a rate of carbon storage.
4. The aquifer management system of claim 1, wherein the set of aquifers are at a single storage site that meets the set of carbon storage requirements.
5. The aquifer management system of claim 1, wherein the set of aquifers are a ranked list of single storage sites that meets the set of carbon storage requirements.
6. The aquifer management system of claim 1, wherein the set of aquifers are multiple aquifers that meet the set of carbon storage requirements.
7. The aquifer management system of claim 1, wherein the machine learning model determines an arrangement of multiple wells in an aquifer that meets the set of carbon storage requirements.
8. The aquifer management system of claim 1, wherein the set of physics-based equations are reservoir engineering equations.
9. The aquifer management system of claim 8, wherein the reservoir engineering equations is selected from at least one of a material balance equation and a deliverability equation, wherein the material balance equation defines relationship between average reservoir pressure and cumulative carbon dioxide injected for the aquifers, and wherein the deliverability equation defines relation between injection rate and difference between average reservoir pressure and wellbore pressure for the aquifers.
10. The aquifer management system of claim 9, wherein the material balance equation is:
$$\bar{p}(t) - p_i = \frac{B_g \int q(t) dt}{Ah\phi C_t}.$$
11. The aquifer management system of claim 9, wherein the deliverability equation is:

$$q(t) = \frac{2}{B_g} \frac{kh}{w} \frac{p_w(t) - \bar{p}(t)}{\ln(\frac{r_e}{r_w}) - 0.75 + S(t)} \cdot$$

12. The aquifer management system of claim 1, wherein the aquifer parameters are selected from at least one of aquifer size, aquifer thickness, aquifer depth, a petrophysical property, a pressure condition, a temperature condition, a fluid property, a solubility condition, a two-phase flow property, a porosity, a fracturing condition, or a permeability.
13. The aquifer management system of claim 1, wherein the simulation parameters are selected from at least one of wellbore pressure, reservoir pressure, injection rate, and cumulative injection.
14. The aquifer management system of claim 1 further comprising: an aquifer selector configured to: select the set of aquifers from a number of candidate aquifers that meet the set of carbon storage requirements to store carbon dioxide using the machine learning model trained using the training dataset.
15. The aquifer management system of claim 14, wherein selecting the set of aquifers from the number of candidate aquifers that meet the set of carbon storage requirements to store carbon dioxide using the machine learning model trained using the training dataset, the aquifer selector is configured to: determine auxiliary quantities for the auxiliary parameters from the input data for the aquifer parameters using the machine learning model; determine values for carbon storage parameters using the auxiliary quantities for the auxiliary parameters; compare the values for the carbon storage parameters with the set of carbon storage requirements to form a comparison; and identify the set of aquifers from the number of candidate aquifers that meet a set of carbon storage requirements based on the comparison.
16. The aquifer management system of claim 14, wherein the aquifer selector is configured to: determine at least one of a range of capacity and injection rate, an injection rate profile, a number of injectors, or a well spacing for the number of the aquifers.
17. The aquifer management system of claim 14, wherein the machine learning model also determines an order in which carbon dioxide is to be stored in the set of aquifers.
18. The aquifer management system of claim 14, wherein the machine learning model also determines a rate at which carbon dioxide is to be stored in the set of aquifers.
19. The aquifer management system of claim 1, wherein the simulation parameters for the aquifers comprise different combinations of aquifer parameters for the aquifers.
20. A computer implemented method for training a machine learning model to select aquifers for carbon storage, the computer implemented method comprising: generating, by a number of processor units, simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models; generating, by the number of processor units, auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters; creating, by the number of processor units, a training dataset using the simulation data and the auxiliary quantities; and training, by the number of processor units, a machine learning model using the training dataset for selecting a set of aquifers that meet a set of carbon storage requirements to store carbon dioxide.
21. The computer implemented method of claim 20, wherein creating, by the number of processor units, a training dataset using the simulation data and the auxiliary quantities comprises: generating, by the number of processors, the training dataset using combinations of data for the aquifer parameters, simulation parameters, and the auxiliary parameters.
22. The computer implemented method of claim 20, wherein the set of carbon storage requirements are selected from at least one of capacity, injection rate, or a rate of carbon storage.
23. The computer implemented method of claim 20, wherein the set of aquifers are at a single storage site that meets the set of carbon storage requirements.
24. The computer implemented method of claim 20, wherein the set of aquifers is a ranked list of single storage sites that meets the set of carbon storage requirements.

25. The computer implemented method of claim 20, wherein the set of aquifers are multiple aquifers that meet the set of carbon storage requirements.
26. The computer implemented method of claim 20, wherein the machine learning model determines an arrangement of multiple wells in an aquifer that meets the set of carbon storage requirements.
27. The computer implemented method of claim 20, wherein the set of physics-based equations are reservoir engineering equations.
28. The computer implemented method of claim 20, wherein the reservoir engineering equations is selected from at least one of a material balance equation and a deliverability equation, wherein the material balance equation defines relationship between average reservoir pressure and cumulative carbon dioxide injected for the aquifers, and wherein the deliverability equation defines relation between injection rate and difference between average reservoir pressure and wellbore pressure for the aquifers.
29. The computer implemented method of claim 28, wherein the material balance equation is:
$$\bar{p}(t) - p_i = \frac{B_g \int q(t) dt}{Ah\phi C_t}.$$
30. The computer implemented method of claim 28, wherein the deliverability equation is:
$$q(t) = \frac{2}{B_g} \frac{kh}{w} \frac{p_w(t) - \bar{p}(t)}{\ln(\frac{r_e}{r_w}) - 0.75 + S(t)}.$$
31. The computer implemented method of claim 20, wherein the aquifer parameters are selected from at least one aquifer size, aquifer thickness, aquifer depth, a petrophysical property, a pressure condition, a temperature condition, a fluid property, a solubility condition, a two-phase flow property, a porosity, a fracturing condition, or a permeability.
32. The computer implemented method of claim 20, wherein the simulation parameters are selected from at least one of wellbore pressure, reservoir pressure, injection rate, and cumulative injection.
33. The computer implemented method of claim 20 further comprising: receiving, by the number of processor units, input data for the aquifer parameters for a number of candidate aquifers; and identifying, by the number of processor units, the set of aquifers that meet the set of carbon storage requirements for carbon storage from the number of candidate aquifers using the machine learning model.
34. The computer implemented method of claim 33, wherein identifying, by the number of processor units, the set of aquifers that meet the set of carbon storage requirements for carbon storage from the number of candidate aquifers using the machine learning model comprises: determining, by the number of processor units, the auxiliary quantities for the auxiliary parameters from the input data for the aquifer parameters using the machine learning model; determining, by the number of processor units, values for carbon storage parameters using the auxiliary quantities for the auxiliary parameters; comparing, by the number of processor units, the values for the carbon storage parameters with the set of carbon storage requirements to form a comparison; and identifying, by the number of processor units, the set of aquifers that meet the set of carbon storage requirements from the number of candidate aquifers based on the comparison.
35. The computer implemented method of claim 34 further comprising: determining, by the number of processor units, at least one of a range of capacity and injection rate, an injection rate profile, a number of injectors, or a well spacing for the number of the aquifers.
36. The computer implemented method of claim 20, wherein the simulation parameters for the aquifers comprise different combinations of aquifer parameters for the aquifers.
37. A computer implemented method for storing carbon dioxide, the computer implemented method comprising: receiving, by a number of processor units, input data for aquifer parameters for a number of candidate aquifers; sending, by the number of processor units, the input data to a machine learning model trained using a training dataset comprising combinations of data for the aquifer parameters, simulation parameters, and auxiliary parameters; receiving, by the number of processor units, results for candidate aquifers from the machine learning model in response to

sending the input data to the machine learning model; and identifying, by the number of processor units, a set of aquifers that meet a set of carbon storage requirements from the number of candidate aquifers based on the results received from the machine learning model.

38. The computer implemented method of claim 37, wherein the results comprise an order in which carbon dioxide is to be stored in the set of aquifers.

39. The computer implemented method of claim 37, wherein the results comprise a rate at which carbon dioxide is to be stored in the set of aquifers.

40. A computer program product for training a machine learning model to select aquifers for carbon storage, the computer program product comprising: a set of one or more computer-readable storage media; program instructions, collectively stored in the set of one or more storage media, cause a number of processor units to perform the following computer operations: generate simulation data for simulation parameters for aquifers using a set of numerical simulation models with historical data for aquifer parameters for the aquifers as inputs to the set of numerical simulation models; generate auxiliary quantities for auxiliary parameters for the aquifers using a set of physics-based equations and the simulation data for a set of selected simulation parameters; create a training dataset using the simulation data and the auxiliary quantities; and train a machine learning model using the training dataset to select a number of the aquifers that meet a set of carbon storage requirements to store carbon dioxide.
